



ENGLISH

COMPREHENSION & VOCABULARY

UPSC / BANK PO / SSC / OTHERS

SESSION - 33

14th March, 2017 @ 12.00 noon

Now to Catch the Sun



Ashok Jhunjhunwala & Bhaskar Ramamurthi

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As the consumer demand for power also varies with time-of-day and season, there is a problem of matching demand to supply, both of which vary independently. One option would be to have excess capacity and get the non-renewable power generators, which are under our control, to back off when needed. However, this strategy has to be adopted judiciously, as it will increase the cost of non-renewable power.

Demand management, where the customer is incentivised to use more power when available and consume less when there is a shortage, will help and will, indeed, become necessary. Smart buildings and factories will take us towards implementation of demand management in time.

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able power to become an unfettered dominant supplier is some kind of large-scale storage. The electric vehicle (EV) is precisely such an application, where the cost of energy per km, including the cost of its storage, has to be only lower than the corresponding cost of a petroleum-based vehicle, to be economically viable.

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But there is another option towards large-scale EV adoption. The key elements of EV technology are available today at the right prices for several types of vehicles widely used in the Indian market. With an innovative, coordinated and market-oriented effort by industry and the government, certain EVs can be produced in India today. Adoption can be rolled out rapidly in a fully market-driven manner.

In these specific segments, India could attain a globally competitive leadership position in three to five years. This effort will simultaneously encourage local intellectual property (IP) generation and the manufacture of most EV subsystems or substantial parts of them. India can move towards substitution of oil imports with locally produced energy and EVs including subsystems.

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Jhunjhunwala is adviser, minister of power, coal, new and renewable energy, and Ramamurthi is director, IIT-Madras



The glasses are half full...

THE ECONOMIC TIMES

7th March, 2017



DISCUSSION ON
14th March (Tuesday)



DISCUSSION ON
15th March (tomorrow)

Canada and beyond

India should keep its trade cards close to its chest in these times of flux

It is a reflection of the state of flux in world trade, post-Trump, that Canada should be making a pitch to expand both trade and investment ties with India. While Canada is the world's tenth largest economy, a member of the G-8 and a major agriculture, food processing and energy powerhouse, its bilateral goods trade with India is a piffling \$7 billion, more or less evenly balanced. Canada's direct investment in India is less than a billion dollars, whereas India's investments in Canada are above \$3 billion. Yet, Canada has said that it would like to expand its footprint in India and seal a trade and investment pact. Its optimism also stems from its recent success in finalising a trade deal with the European Union, coinciding with the virtual demise of the US-EU Trans Atlantic Trade and Investment Partnership. Under this, the EU will eliminate tariffs on most of Canada's exports, while Canada in exchange will offer more free access for people, obviously 'trumping' the US here. However, it has its own share of headaches to deal with, emanating from across the border. The US is Canada's largest trading partner, their mutual merchandise trade amounting to \$575 billion, with a trade deficit of \$5 billion in Canada's favour; in the case of services, the US holds the advantage in about \$90 billion bilateral trade. But there is a cloud of uncertainty arising out of the Trump administration's stated 'trade policy agenda', which does not take a kind view of NAFTA. Canada is obviously exploring options in this context.

Canada is playing a hard dealmaker, pushing for international arbitration platforms. There is no need for India to rush into a knee-jerk response. Its model Bilateral Investment Treaty makes its policy transparent for the stakeholders concerned. The Centre should adopt a consultative approach so that the mistakes of earlier FTAs are not repeated. Rather, the focus should be on enhancing ease of doing business, and trade facilitation measures as agreed to in the Bali Ministerial of the WTO. India also needs to work out its negotiating strategy in RCEP and the WTO. The US trade agenda warns that "The Trump Administration will use all (emphasis added) possible leverage to encourage other countries to give US producers fair, reciprocal access to their markets." Simply put, this could translate into pressure to dilute intellectual property rights and eliminate, as opposed to reduce, tariffs. Barriers to trade in digital data and services are an area of interest to the US. With TPP no longer in the picture, India's negotiating position in RCEP could improve. It should not flinch from seeking market access in services in exchange for more free tariff lines.

Global trade is a win-win game in principle, provided the actors play by the rules. Unless the political and economic contours of a new trade regime become clearer, India should not make precipitate moves on the world trade chessboard.



THE HINDU
BusinessLine

9th March, 2017

Now to Catch the Sun



Ashok Jhunjhunwala & Bhaskar Ramamurthi

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The glasses are half full...

FILE PHOTO

SOLAR ENERGY Now to Catch the Sun

THE ECONOMIC TIMES

7th March, 2017

SOLAR ENERGY - Now to Catch the Sun

- The recent tenders for setting up and operating solar and wind power generation farms show that the price of renewable power is now approaching that from **greenfield** coal-based power plants in India. It will not take too long for solar **photovoltaics** to become the cheapest source of power. But solar power is generated only during the day , that too **intermittently**. Similarly , wind power is generated during certain months and the power output also varies. If energy storage was inexpensive, one could have dealt with this intermittency by storing excess energy generated to be used when needed.
- Since **grid-connected** energy storage continues to be very expensive, generated power needs to be consumed instantly. As the consumer demand for power also varies with time-of-day and season, there is a problem of matching demand to supply , both of which vary independently . One option would be to have excess capacity and get the non-renewable power generators, which are under our control, to back off when needed. However, this strategy has to be adopted **judiciously** , as it will increase the cost of non-renewable power.
- **Demand management**, where the customer is **incentivised** to use more power when available and consume less when there is a shortage, will help and will, **indeed**, become necessary . Smart buildings and factories will take us towards implementation of demand management in time.
- But what would really enable renewable power to become an **unfettered** dominant supplier is some kind of large-scale storage. The electric vehicle (EV) is precisely such an application, where the cost of energy per km, including the cost of its storage, has to be only lower than the corresponding cost of a petroleum-based vehicle, to be economically **viable**.
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Greenfield Project

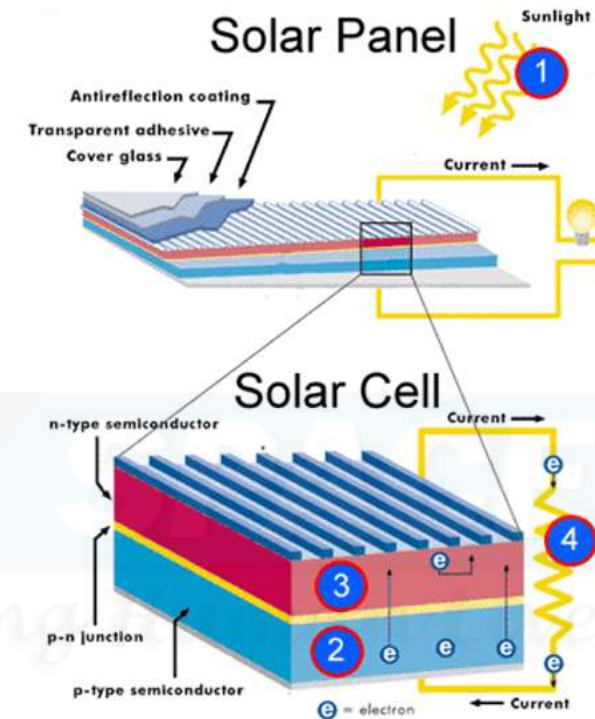


Brownfield Project



Photovoltaics

*Typical **photovoltaic** system uses solar panels. It directly converts sunlight to electricity without any moving parts*



Photovoltaics

Conversion of light into electricity using semiconducting materials that exhibit the **photovoltaic** effect.

Intermittently

There is *intermittent* rain
for the past three days.



Intermittently

Not happening regularly or continuously

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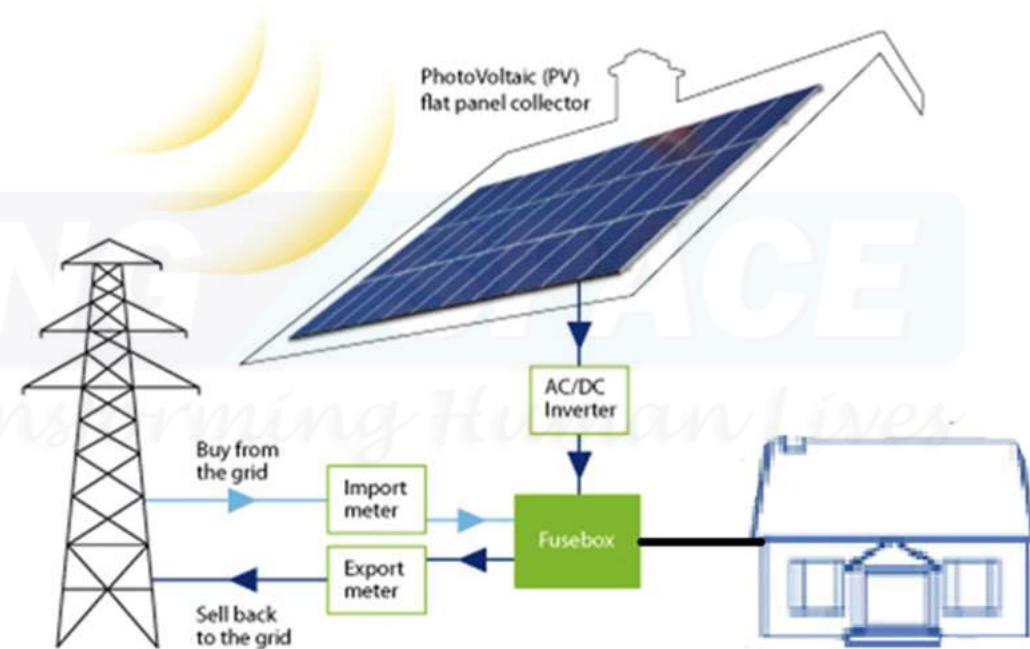
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Grid connected photovoltaic power system

- Electricity generated from solar photovoltaic power system is connected to the utility grid.
- It supplies the excess power, beyond consumption to the utility grid.



Judiciously

*As the resources available
to us are scarce,
government should make
judicious use of the
resources.*



सत्यमेव जयते

Judiciously

Using reason and good judgment in making decisions

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DEMAND MANAGEMENT

- It is the planning methodology used to predict, plan and manage the demand for products and services.
- This can be done at macro levels as in economics and at micro levels within individual organizations.

Incentive

Cash *incentives* will go a long way in increasing digital transactions.



Incentive

Something that encourages a person or organization to do something

Indeed

Indeed, Paris Pact is one of the most important multilateral treaties in the world



PARIS2015
UN CLIMATE CHANGE CONFERENCE
COP21·CMP11

I am happy, indeed proud, as my son got selected for SBI PO



Indeed

Really / certainly, often used to emphasize something

Indeed

Add some extra information

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Unfettered

*India should devise public health policy with critical review of the scientific evidence with open minds **unfettered** by customs and dogma.*

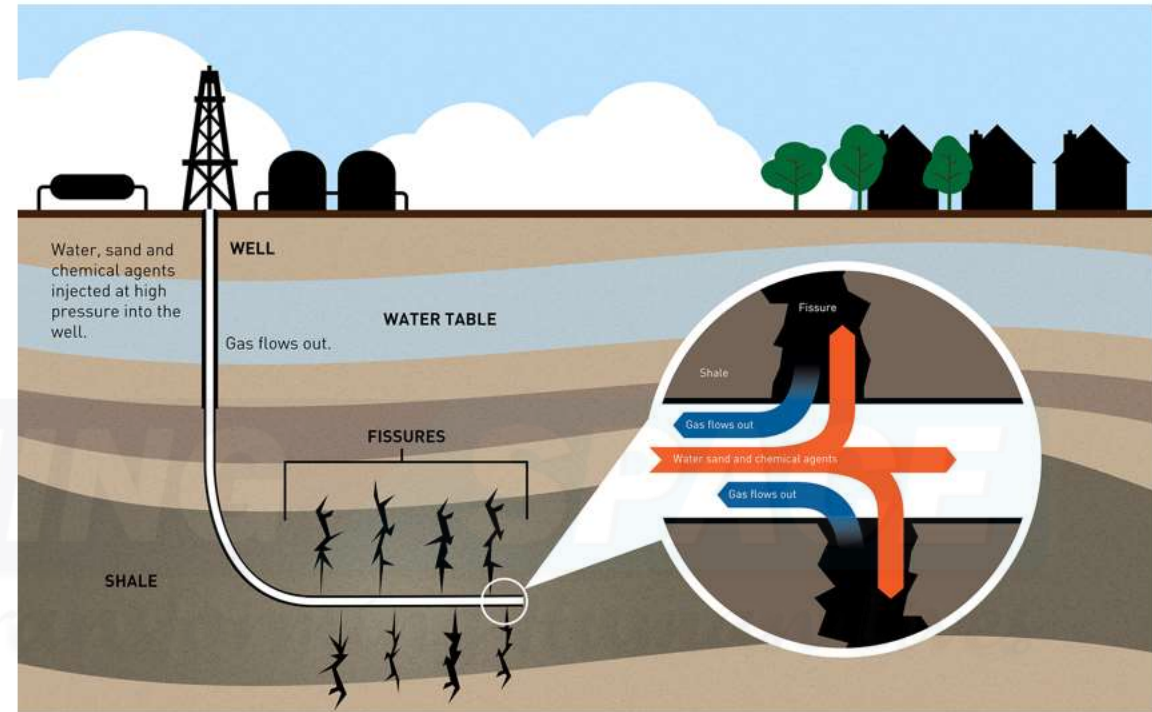


Unfettered

Not limited by rules or any other controlling influence

Viabile

*When the oil prices goes below \$ 50 per barrel, then extraction of shale oil becomes economically **unviable***

**Viabile**

Likely to succeed / able to work as required / able to succeed

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Overcome

*She **overcame** adversity, worked hard and came back much more stronger after she lost one hand in the road accident.*



Overcome

Succeed in dealing with a problem or difficulty

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Imply

*Rajesh asked for increasing the fresh investment in the company and Arun's silence **implies** his consent.*



Imply

To communicate an idea or feeling without saying it directly

Nevertheless



*I know something about
Bitcoins, but his talk on the
subject was interesting
nevertheless*



*The apartment complex is well
guarded, but robberies occur
nevertheless*

Nevertheless

Despite what has just been said or referred to

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Simultaneously



Simultaneously

Happening or being done at exactly the same time

Concurrently

*The two super hit movies are being telecast **concurrently** by rival TV channels.*



Concurrently

Happening or existing at the same time

Intellectual Property (IP)

INTELLECTUAL PROPERTY (IP)



It is the creations of the mind, such as inventions, literary and artistic works, designs, symbols, names and images used in commerce. It is protected by patents, copyrights & trademarks.

1. PATENT

It is exclusive right granted for an invention, may be a product or a process, a new way of doing something normally valid for 20 years.

2. COPYRIGHT

Right of the creators over literary and artistic works may be books, paintings, films, sculptures, databases, advertisements etc.

3. TRADEMARK

It is a sign of distinguishing the goods or services of one Enterprise from those of others

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LIVE

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Intrinsic



Diamonds have got high **intrinsic** value



Geography is the **intrinsic** part of the school curriculum



Paper currency has got no **intrinsic** value

Intrinsic

Essential nature of a thing / basic characteristics and extremely important

Slew

*None of the batsmen could reach a figure of 30 runs and a **slew** of single digit scores resulted in the defeat of Australia in the second test against India.*



Slew

A large number of quantity of something

Single-minded

Corporate hospitals in our country are working with the **single-minded** pursuit of profit without caring for the welfare of the patients.



Single-minded

Concentrating on only one aim.

Pursue

*Raju, the heavily indebted person, was being **pursued** by the creditors.*



Pursue

Follow or chase someone or something

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Laissez-faire

*The company was into problems before he became CEO and the situation further worsened with his **laissez-faire** attitude.*



Laissez-faire

Unwillingness to get involved in or influence other people's activities

Laissez-faire

The *laissez-faire* policy has dominated American politics during the 2nd half of 20th century.



Laissez-faire

This is the idea that the businesses should be free to develop without the involvement or control of government

Indigenous

Thanks to green revolution of 1960s, we became self sufficient with **indigenous** production of food grains



Indigenous

Naturally existing in a place / country

Economies of scale

500 cards



Each costs Rs. 10

5,000 cards



Each costs Rs. 8

Economies of scale

Cost advantage that arises with increased output of a product

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DISCUSSION ON 15th Mar (tomorrow)

THE SAGE HINDU
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LIVE



LEARNING SPACE

Transforming Human Lives

WE EXPRESS OUR
SINCERE THANKS
FOR VIEWING THIS VIDEO



Presented by



LEARNING SPACE

...Transforming Human Lives



OUR TEAM

G. V. Rao
D. Sunil Kumar
Y. Sri Poojita

G. Ravi
D. Joji
K. Victor

For Suggestions:
suggestions@learning

To Contact us:
info@learningspace.in

Visit us at:
www.learningspace.in

0866-24 44472
098499 42299